

Team Name

Vector Nexus

Project Name

PathPilot AI

Tagline

AI-Powered Career Competency Navigation System

Problem Statement

In 2026, students and aspiring software engineers face a major challenge in understanding what skills they truly need for industry roles. Most learners consume random tutorials, courses, and certifications without having a clear roadmap or understanding of industry expectations.

Current platforms provide generic learning paths but fail to:

- analyze individual competency levels
- identify exact skill gaps
- provide role-specific guidance
- align learning with real-world hiring trends
- integrate personal documents, resumes, and learning resources

As a result, many students struggle with:

- lack of direction
- inconsistent learning
- poor project selection
- interview unpreparedness
- skill mismatch with job requirements

Proposed Solution

PathPilot AI is an intelligent AI-driven career competency navigation system that helps users understand:

- where they currently stand
- what skills they are missing
- what industry expects
- what to learn next
- how to become job-ready systematically

The platform uses:

- Retrieval-Augmented Generation (RAG)
- Local LLMs using Ollama
- Vector embeddings
- AI workflow orchestration
- Real-time industry trend analysis

to generate personalized career guidance.

Core Features

1. Resume & Skill Analysis

Users can upload:

- resumes
- PDFs
- notes
- certificates
- GitHub project documentation

The AI extracts:

- technical skills
- project experience
- competency levels
- learning patterns

and generates detailed insights.

2. Skill Gap Analysis

The system compares user skills with industry expectations for roles such as:

- SDE-1
- Backend Developer

- AI Engineer
- Data Engineer
- Full Stack Developer

Output includes:

- missing skills
- competency gap percentage
- improvement recommendations

3. Personalized Learning Roadmap

The platform creates structured learning paths based on:

- user goals
- current skill level
- role requirements
- trending technologies

Roadmaps include:

- weekly plans
- recommended topics
- mini projects
- interview preparation checkpoints

4. AI Knowledge Assistant (RAG System)

Users can ask questions from uploaded documents and knowledge bases such as:

- interview notes
- DSA materials
- AI/ML resources
- system design documents

The assistant provides contextual answers using vector retrieval and local LLM inference.

5. Industry Trend Intelligence

Using web search and RSS feeds, the platform tracks:

- trending technologies
- hiring skills
- emerging AI tools
- industry demand

This keeps recommendations dynamic and future-oriented.

6. Career Readiness Score

The system generates visual analytics for:

- DSA readiness
- project quality

- backend competency
- AI knowledge
- interview preparedness

and calculates an overall industry readiness score.

Technology Stack

Core AI Framework

- LangChain
- LangGraph
- LangChain Community
- LangChain Ollama
- LangChain Chroma
- LangChain HuggingFace

Frontend

- Streamlit

Vector Database & Embeddings

- ChromaDB
- Sentence Transformers
- ONNX Runtime

PDF Processing

- PyPDF

Search & Intelligence

- DuckDuckGo Search
- FeedParser

Visualization

- Plotly

Utilities

- Python Dotenv
- Bleach
- XHTML2PDF

Local LLM Runtime

- Ollama

Architecture Overview

User Uploads Documents



PDF Processing & Text Extraction



Embedding Generation



ChromaDB Vector Storage



Retriever + LangChain



Ollama Local LLM



LangGraph Workflow Orchestration



AI Recommendations & Career Insights



Streamlit Dashboard UI

Expected Outcomes

PathPilot AI aims to:

- improve career clarity
- reduce random learning
- provide structured guidance
- increase interview readiness
- help users align with real-world industry expectations

Innovation Aspect

Unlike generic AI chatbots or simple PDF assistants, PathPilot AI acts as:

- a career intelligence platform
- a competency analysis system
- a personalized AI mentor
- a roadmap generator

The project focuses on practical industry alignment instead of only conversational AI.

Future Scope

Future enhancements may include:

- GitHub repository analysis
- LeetCode progress tracking
- AI mock interviews
- voice-based AI mentor
- placement analytics dashboard
- collaborative team skill analysis
- recruiter evaluation mode

Why This Project Matters

PathPilot AI addresses a real-world problem faced by millions of students and developers: understanding what to learn, why to learn it, and how to become industry-ready efficiently.

The project combines:

- AI engineering
- RAG systems
- vector databases
- workflow orchestration
- analytics dashboards
- local LLM deployment

into a practical and scalable solution.